



STM Exercise 6

Phase Diagrams

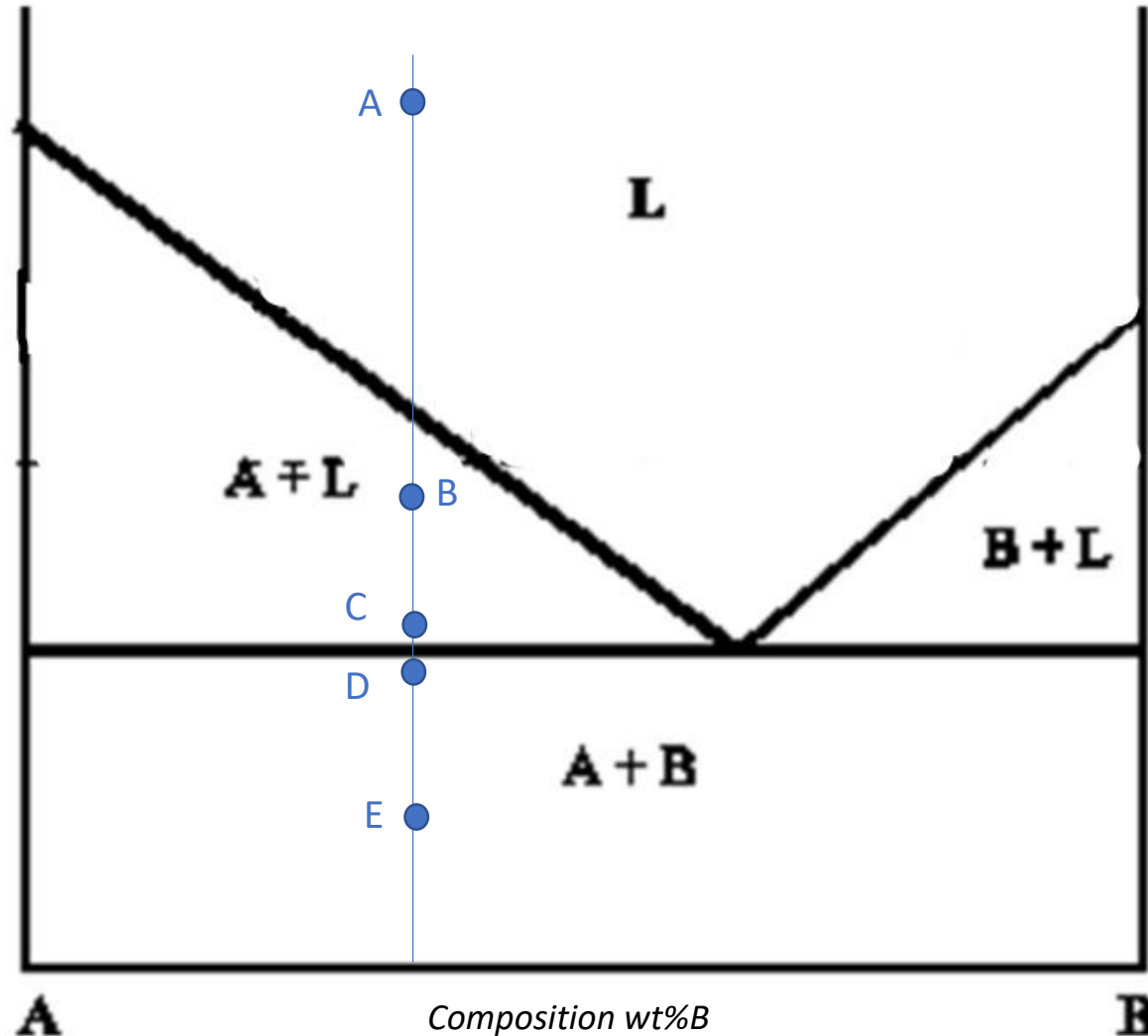
(no solubility in the solid state)

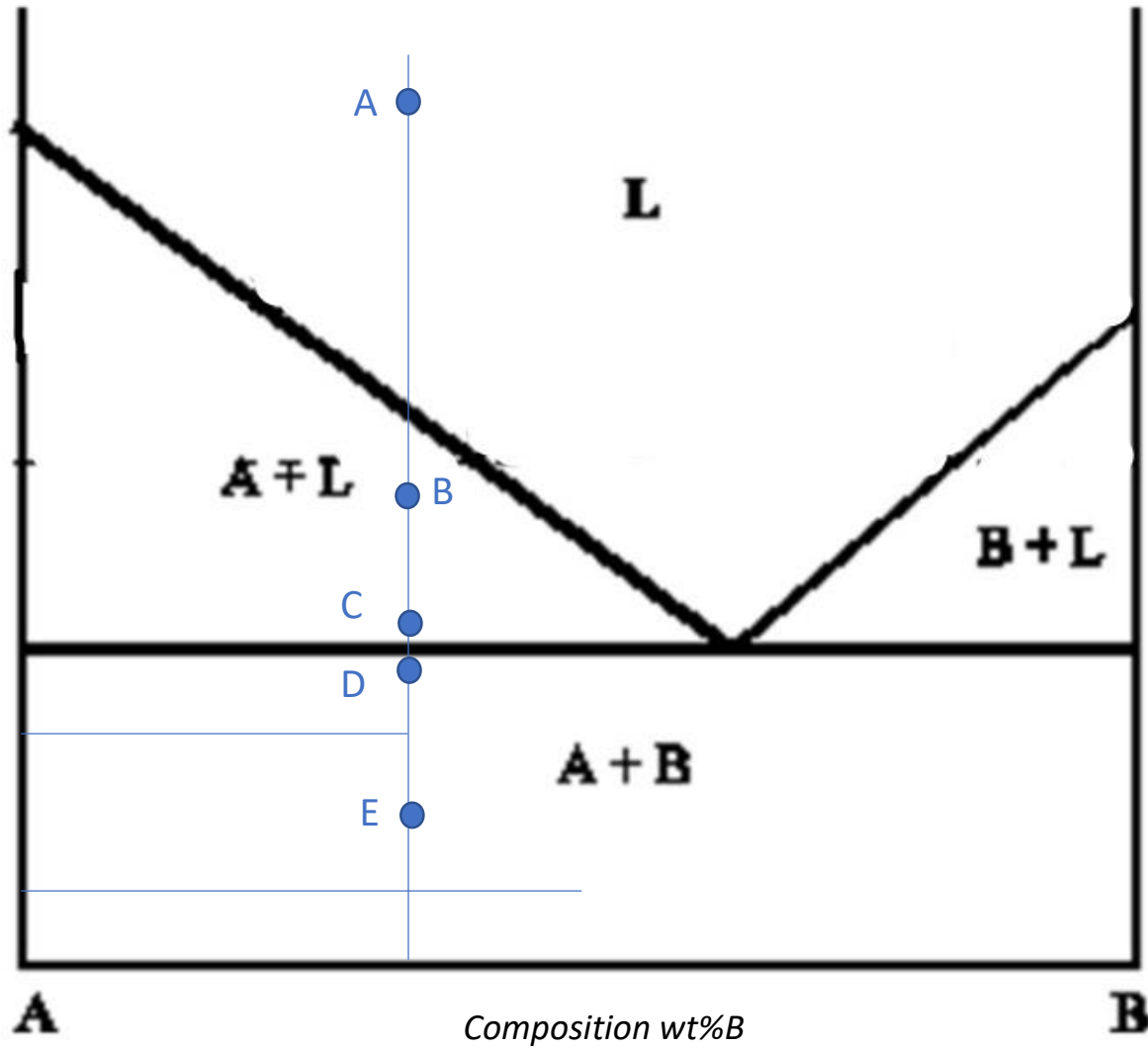
(partial solubility in the solid state)

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Exercise 1: determine the phase present, their composition and relative amounts in the point **A**, **B**, **C** (at the beginning of the eutectic transformation), **D** (at the end of the eutectic transformation) and **E** indicated in the phase diagram. Schematize the microstructure in each point.



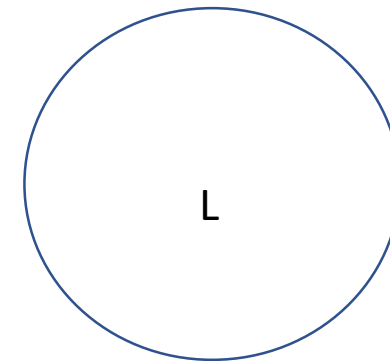


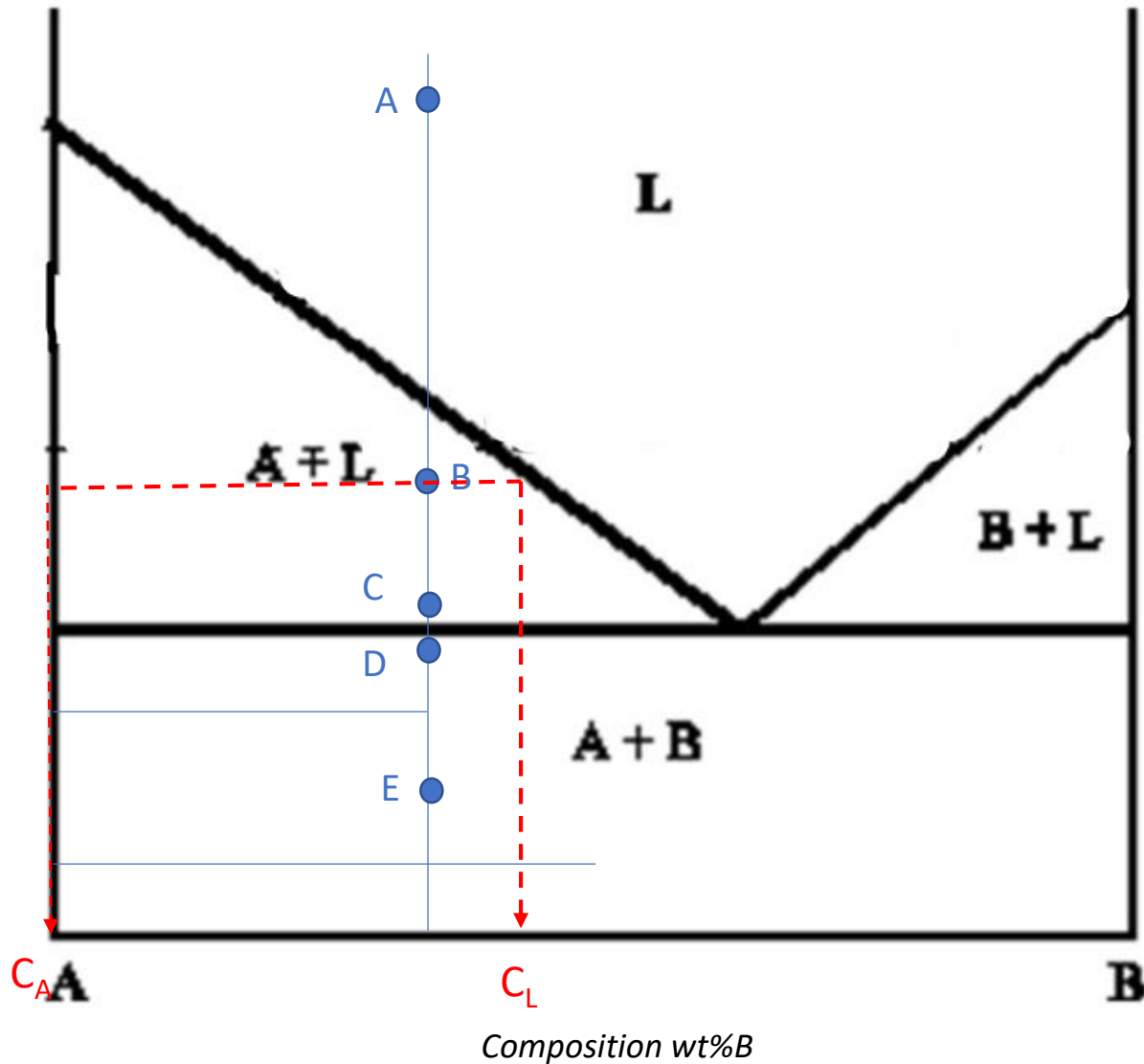
Point A:

Phases: 1 phase liquid

Composition: 35%B, 65%A

Relative amount: 100% liquid phase





Point B:

Phases: 2 phases solid (A) + liquid (L)

Composition: 2 phases region → horizontal rule

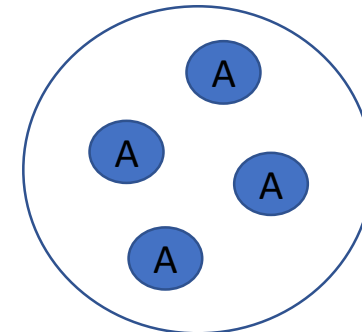
Composition of liquid (L): $C_L = 45\% \text{ B}, 55\% \text{ A}$

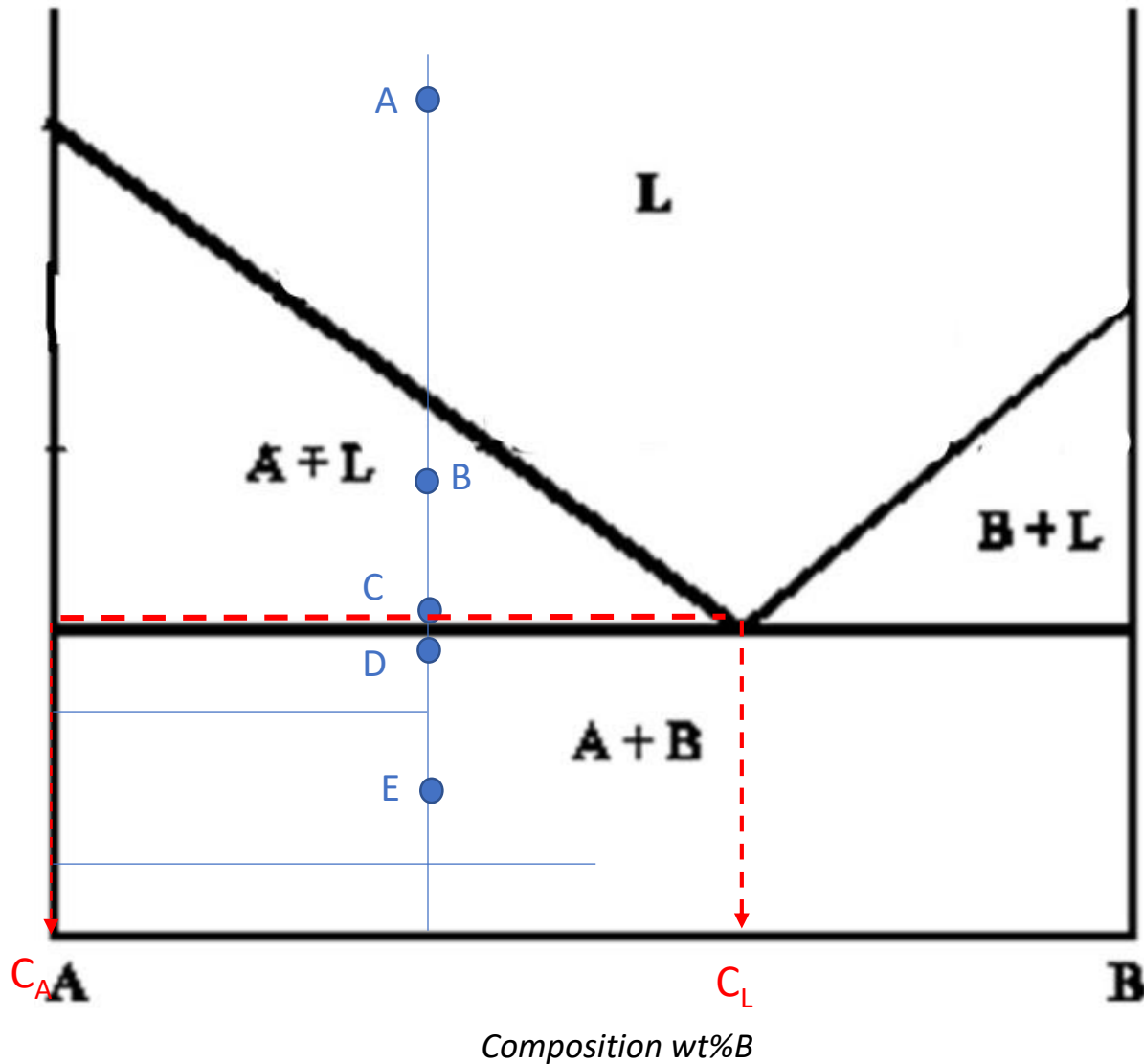
Composition of solid (A): $C_A = 100\% \text{ A}$

Relative amount: 2 phases region → lever rule

$$\% \text{ A} = (45 - 35) / (45 - 0) * 100 = 22\%$$

$$\% \text{ L} = (35 - 0) / (45 - 0) * 100 = 78\%$$





Point C:

Phases: 2 phases solid (A) + liquid (L)

Composition: 2 phases region → horizontal rule

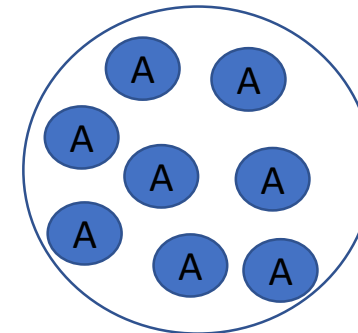
Composition of liquid (L): $C_L = 60\% \text{ B}, 40\% \text{ A}$

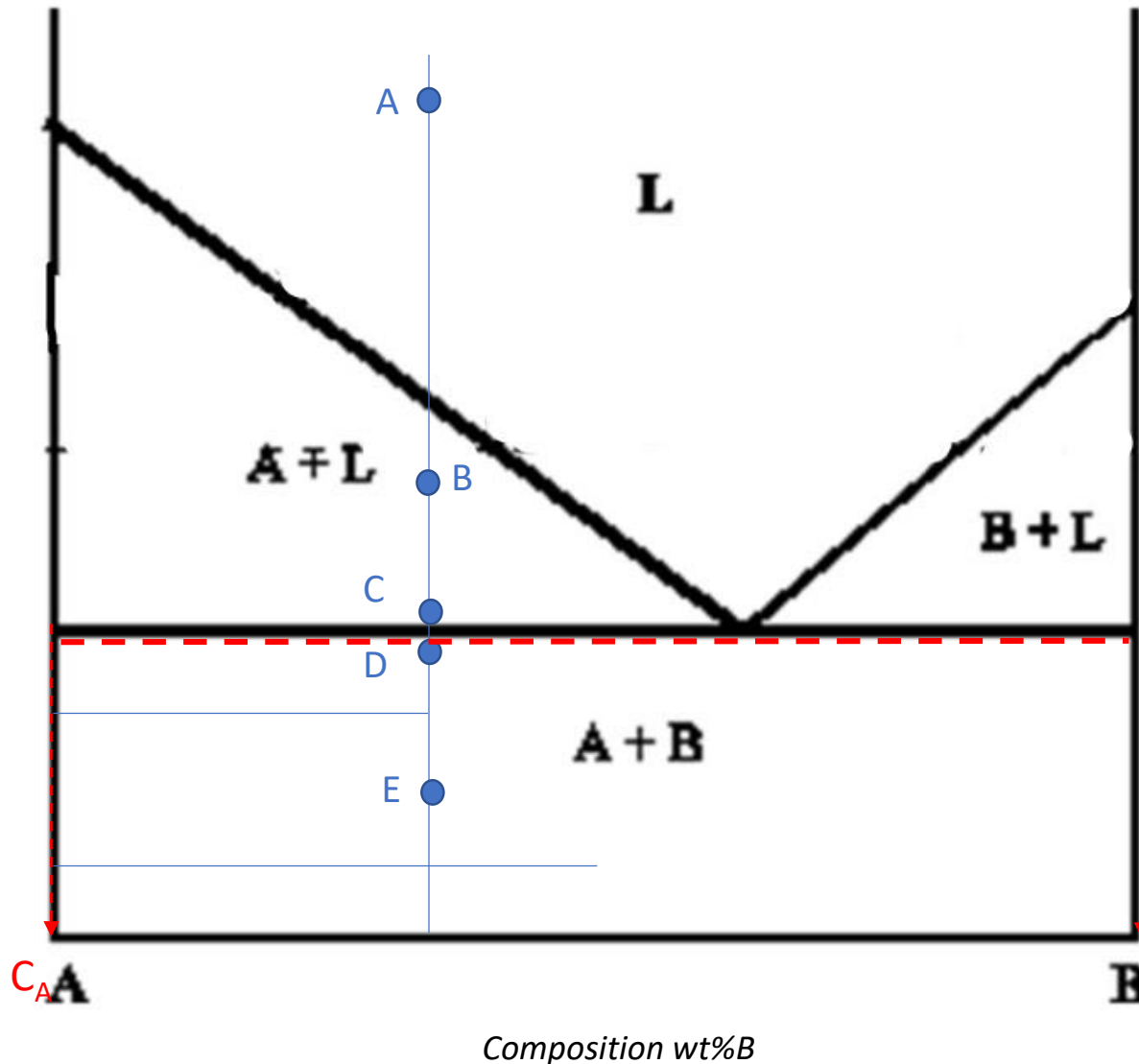
Composition of solid (A): $C_A = 100\% \text{ A}$

Relative amount: 2 phases region → lever rule

$$\% \text{ A} = (60 - 35) / (60 - 0) * 100 = 42\%$$

$$\% \text{ L} = (35 - 0) / (60 - 0) * 100 = 58\%$$





Point D:

Phases: 2 phases solid (A) + solid (B)

Composition: 2 phases region →
horizontal rule

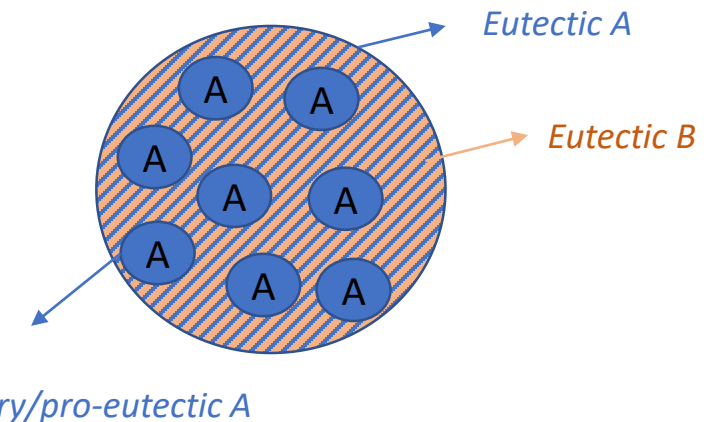
Composition of Solid (A): $C_A = 100\%A$

Composition of solid (B): $C_B = 100\%B$

Relative amount: 2 phases region →
lever rule

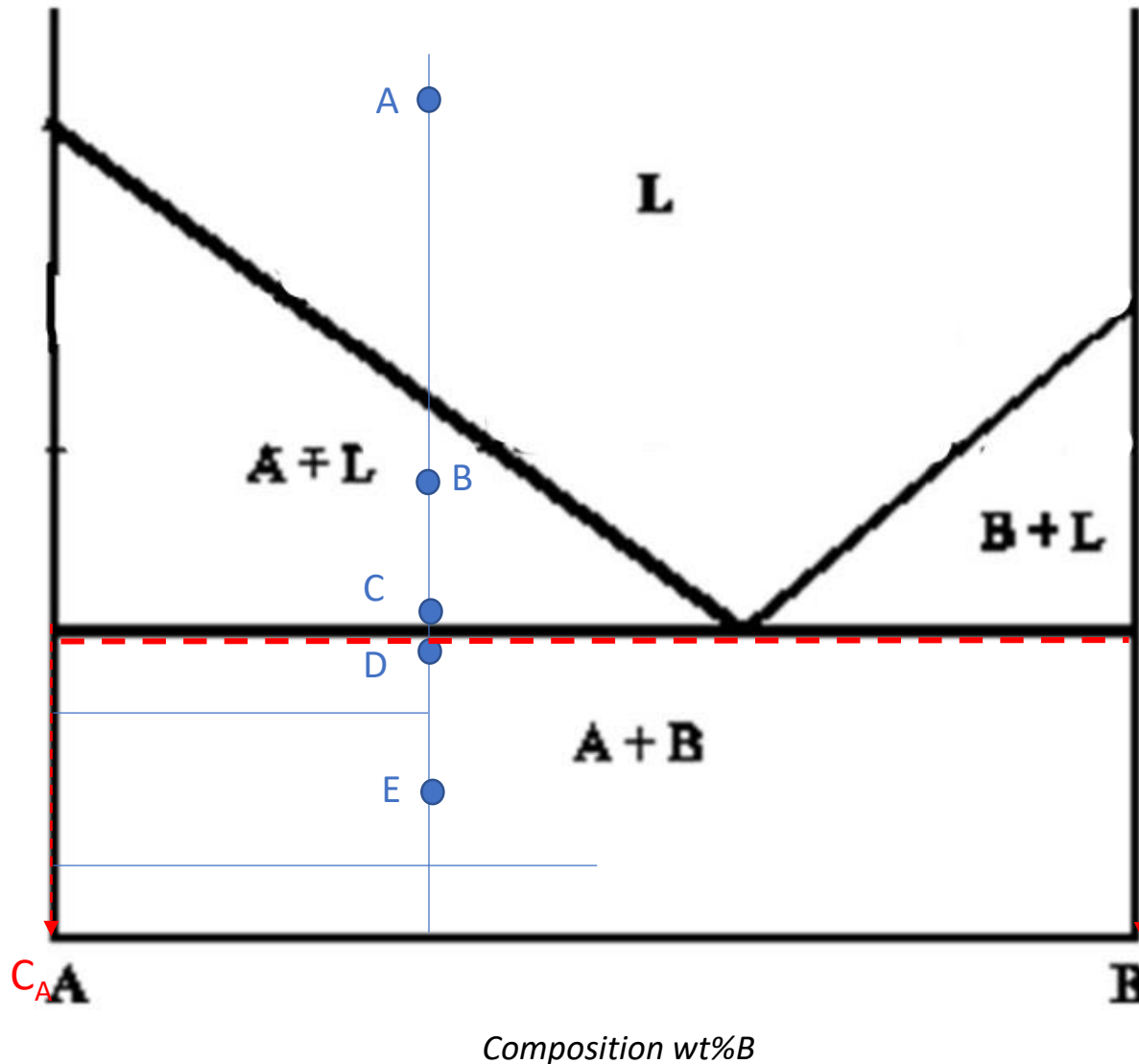
$$\% A_{TOT} = (100-35)/(100-0)*100=65\%$$

$$\% B = (35-0)/(100-0)*100=35\%$$



Primary/pro-eutectic A (As calculated in C, no more formed later): $\% A = (60-35)/(60-0)*100=42\%$

$$\% A_{EUT} = 65-42=23\%$$



Point E:

Phases: 2 phases solid (A) + solid (B)

Composition: 2 phases region →
horizontal rule

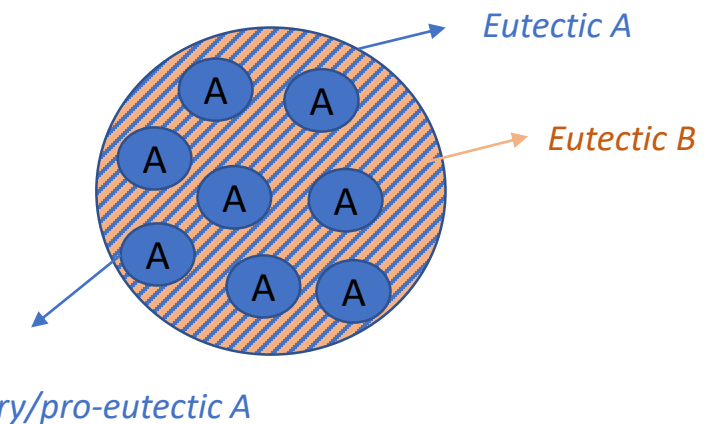
Composition of Solid (A): $C_A = 100\%A$

Composition of solid (B): $C_B = 100\%B$

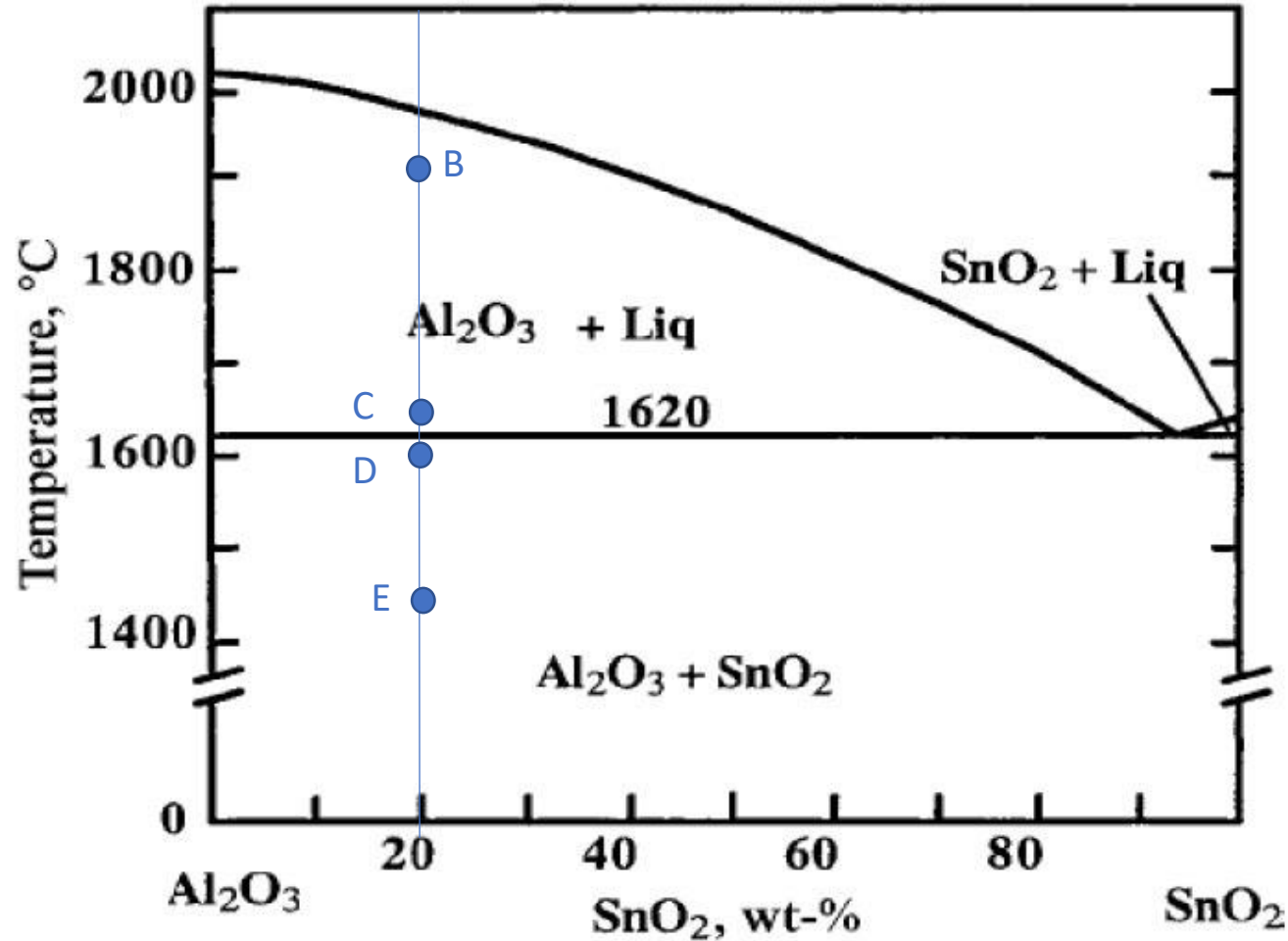
Relative amount: 2 phases region →
lever rule

$$\%A = (100-35)/(100-0)*100=65\%$$

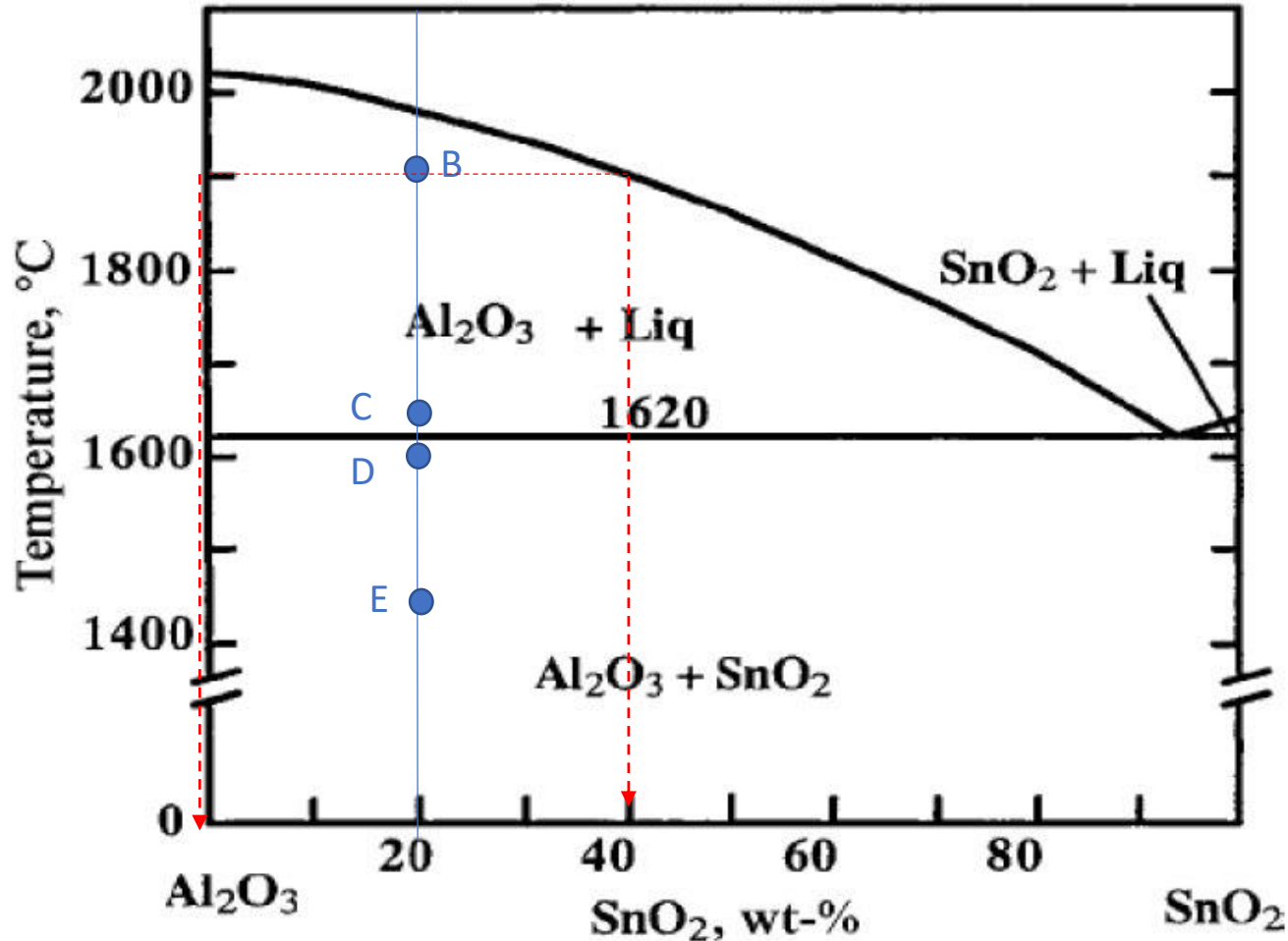
$$\%B = (35-0)/(100-0)*100=35\%$$



Exercise 2: determine the phase present, their composition and relative amounts in the points **B**, **C** (at the beginning of the eutectic transformation), **D** (at the end of the eutectic transformation) and **E** indicated in the phase diagram. Schematize the microstructure in each point.



Exercise 2: determine the phase present, their composition and relative amounts in the points B, C (at the beginning of the eutectic transformation), D (at the end of the eutectic transformation) and E indicated in the phase diagram. Schematize the microstructure in each point.



Point B:

Phases: 2 phases solid (Al_2O_3) + liquid (L)

Composition: 2 phases region $\rightarrow$ horizontal rule

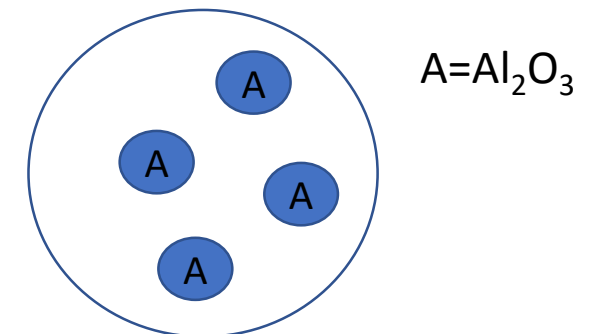
Composition of liquid (L): = 40% SnO_2 , 60% Al_2O_3

Composition of solid: = 100% Al_2O_3

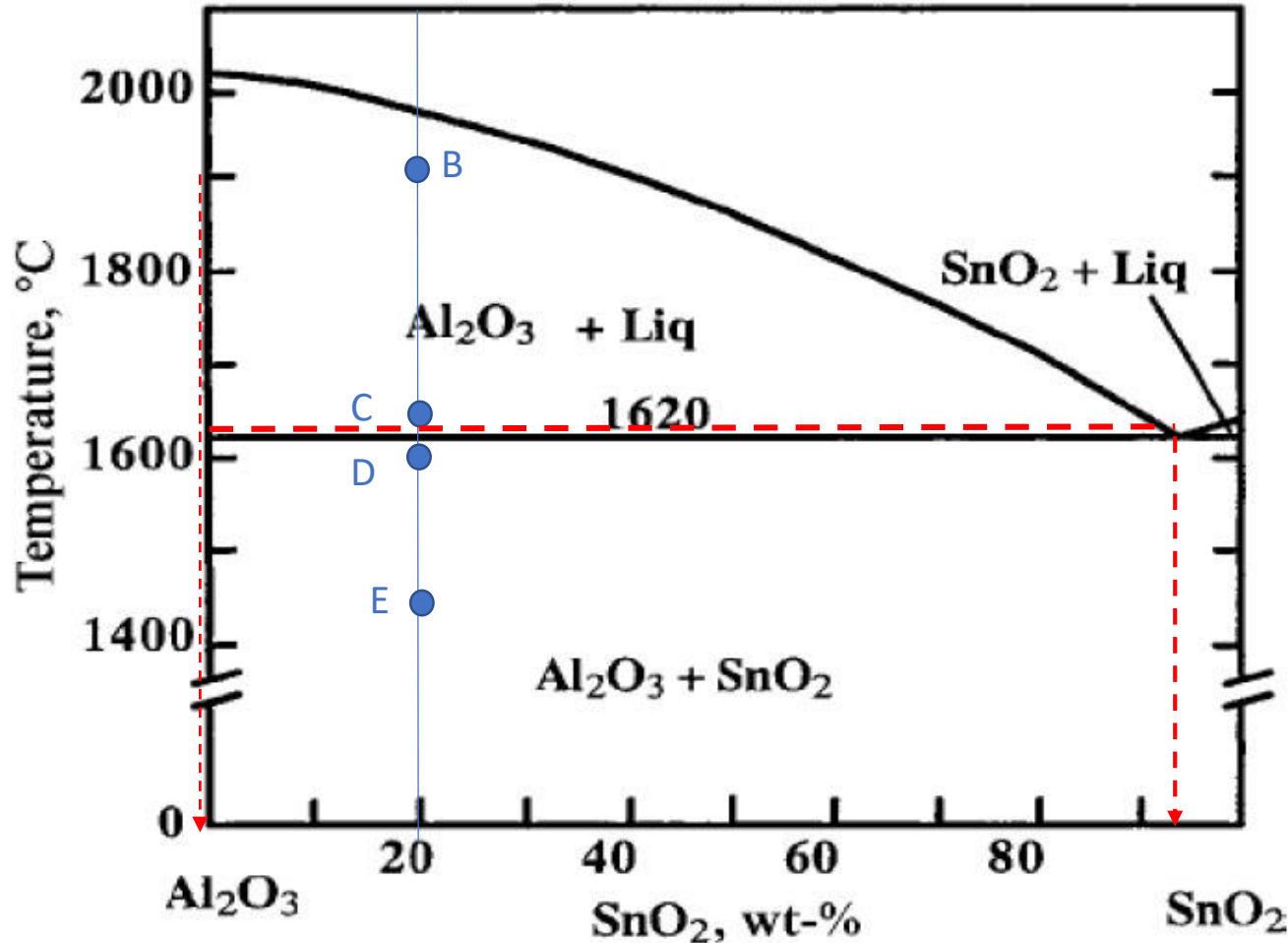
Relative amount: 2 phases region $\rightarrow$ lever rule

$$\% \text{Al}_2\text{O}_3 = (40 - 20) / (40 - 0) * 100 = 50\%$$

$$\% \text{L} = (20 - 0) / (40 - 0) * 100 = 50\%$$



Exercise 2: determine the phase present, their composition and relative amounts in the points **B**, **C** (at the beginning of the eutectic transformation), **D** (at the end of the eutectic transformation) and **E** indicated in the phase diagram. Schematize the microstructure in each point.



Point C:

Phases: 2 phases solid (Al₂O₃) + liquid (L)

Composition: 2 phases region → horizontal rule

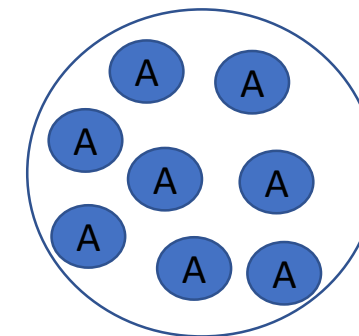
Composition of liquid (L): = 92% SnO₂, 8% Al₂O₃

Composition of solid: = 100% Al₂O₃

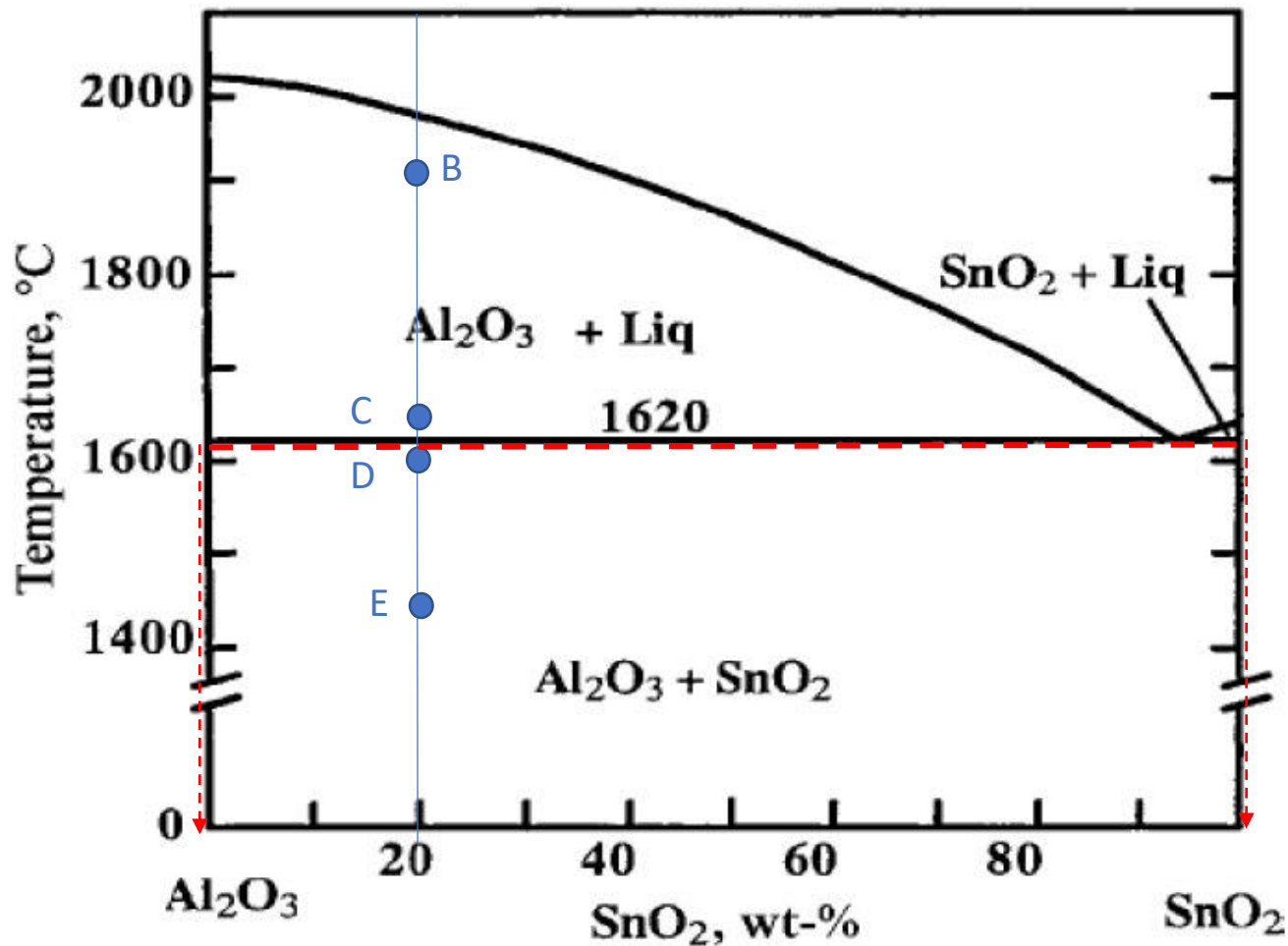
Relative amount: 2 phases region → lever rule

$$\% \text{Al}_2\text{O}_3 = (92 - 20) / (92 - 0) * 100 = 78\%$$

$$\% \text{L} = (20 - 0) / (92 - 0) * 100 = 22\%$$



A = Al₂O₃



Point D:

Phases: 2 phases solid (Al_2O_3) + solid (SnO_2)

Composition: 2 phases region → horizontal rule

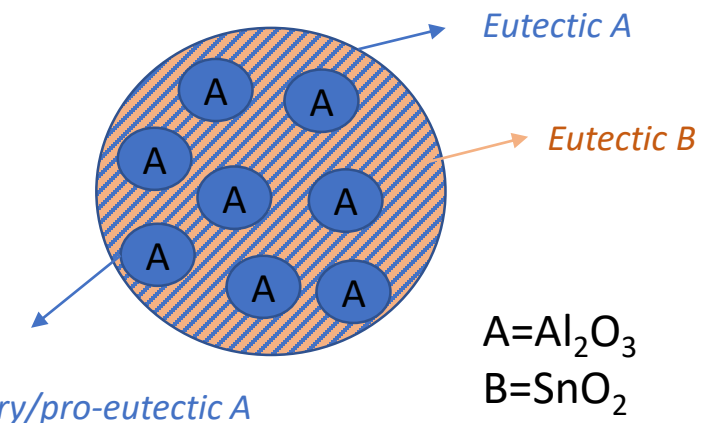
Composition of Solid A = 100% Al_2O_3

Composition of solid (B): $C_B = 100\%\text{SnO}_2$

Relative amount: 2 phases region → lever rule

$\% \text{Al}_2\text{O}_3_{\text{TOT}} = (100-20)/(100-0)*100=80\%$

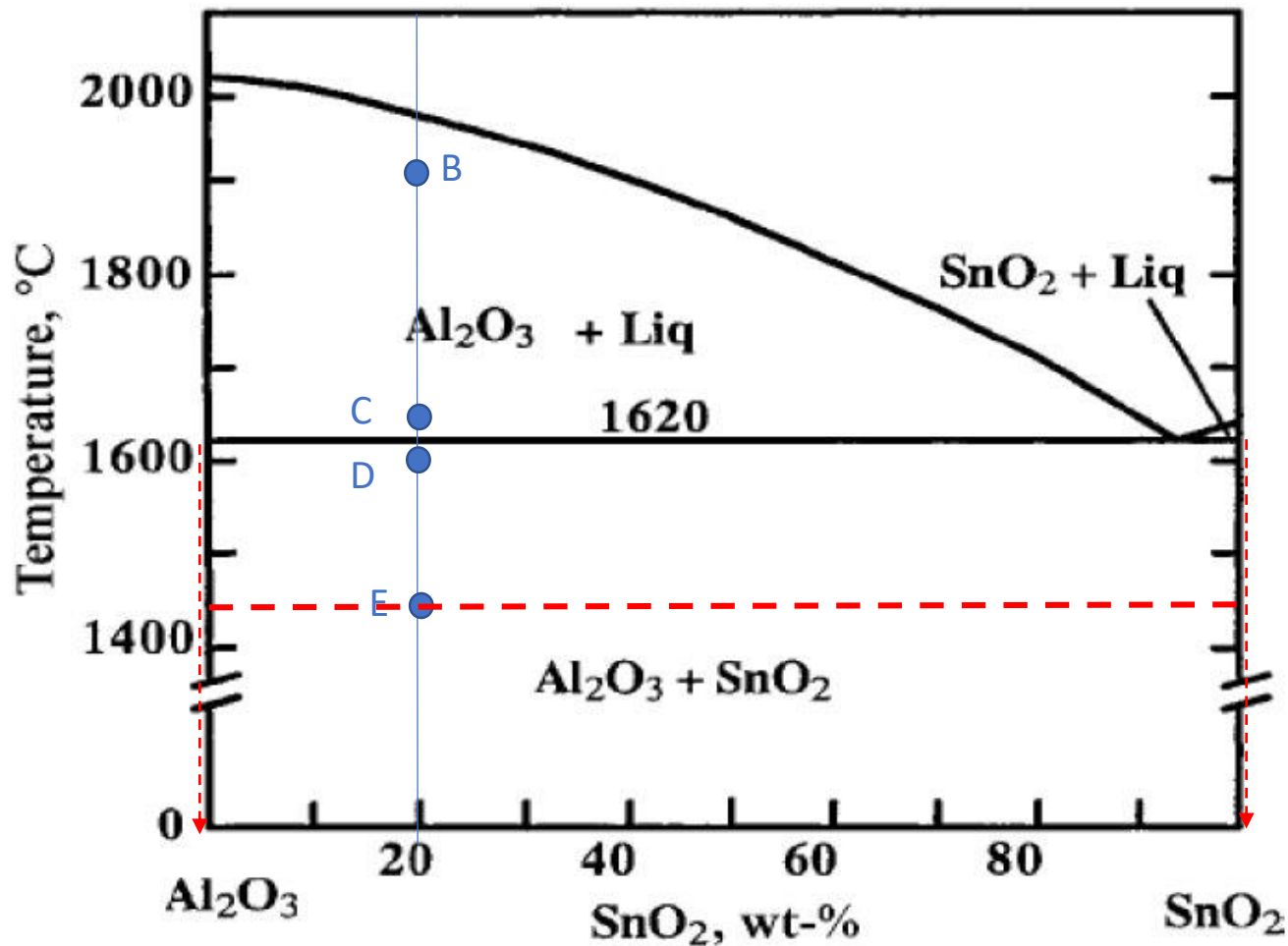
$\% \text{SnO}_2 = (20-0)/(100-0)*100=20\%$



Primary/pro-eutectic Al_2O_3 (As calculated in C, no more formed later): %

$\text{Al}_2\text{O}_3_{\text{PRIM}} = (92-20)/(92-0)*100=78\%$

$\%A_{\text{EUT}} = 80-78=2\%$



Point E:

Phases: 2 phases solid (Al_2O_3) + solid (SnO_2)

Composition: 2 phases region $\rightarrow$ horizontal rule

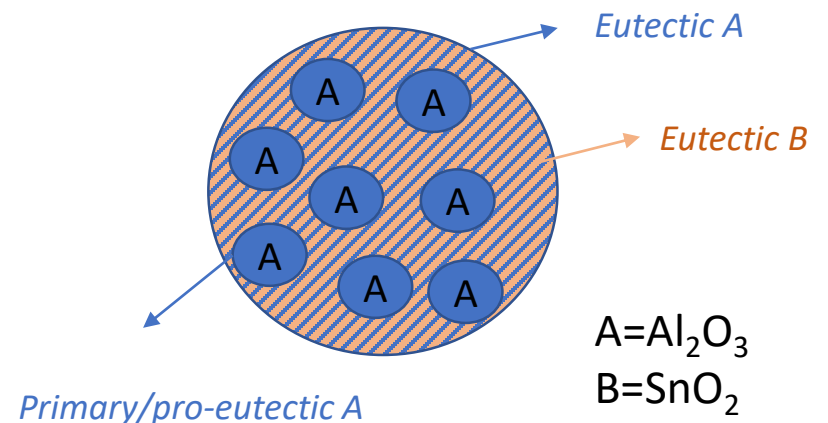
Composition of Solid A = $100\%\text{Al}_2\text{O}_3$

Composition of solid (B): $C_B = 100\%\text{SnO}_2$

Relative amount: 2 phases region $\rightarrow$ lever rule

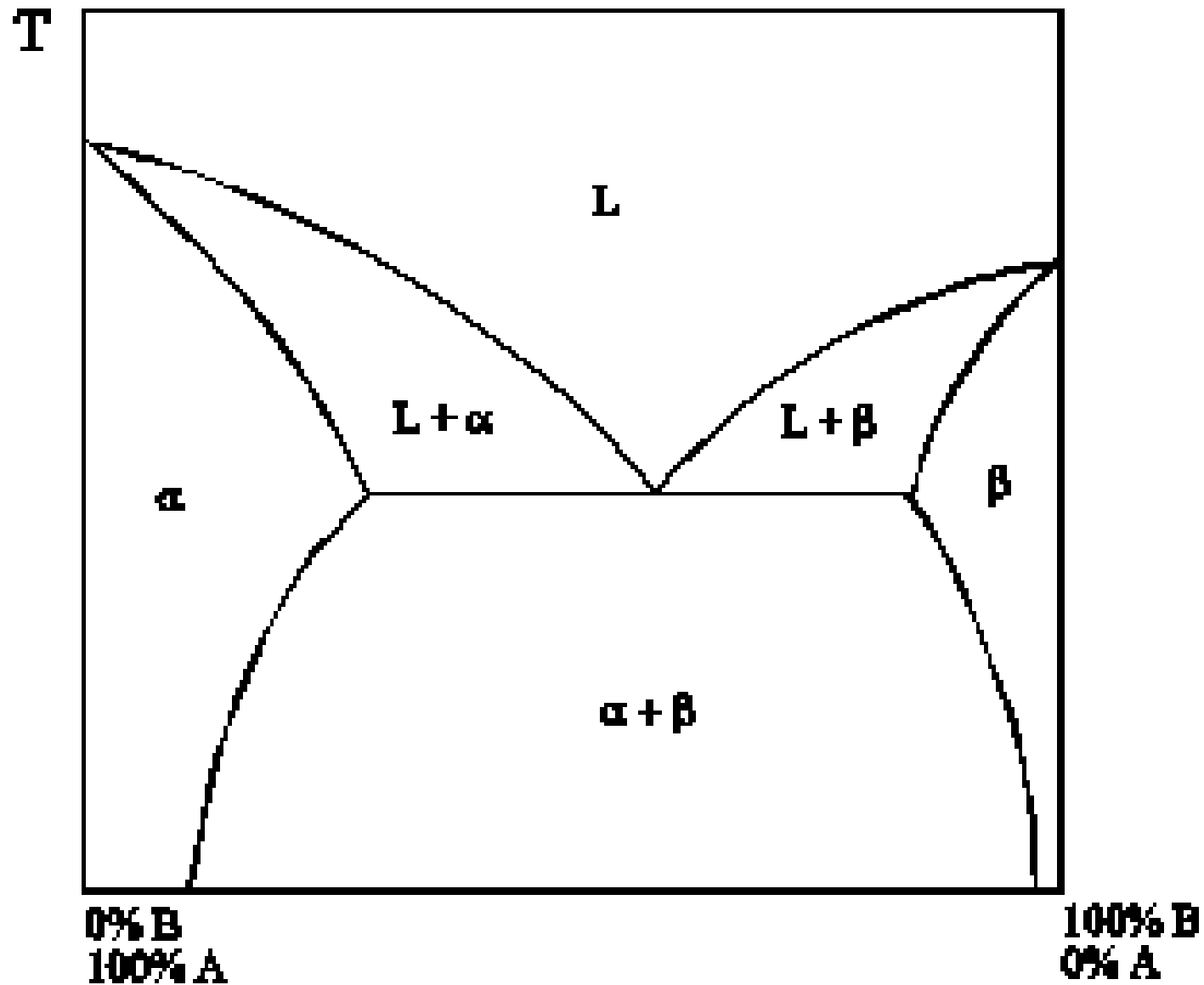
$$\% \text{Al}_2\text{O}_3_{\text{TOT}} = (100 - 20) / (100 - 0) * 100 = 80\%$$

$$\% \text{SnO}_2 = (20 - 0) / (100 - 0) * 100 = 20\%$$

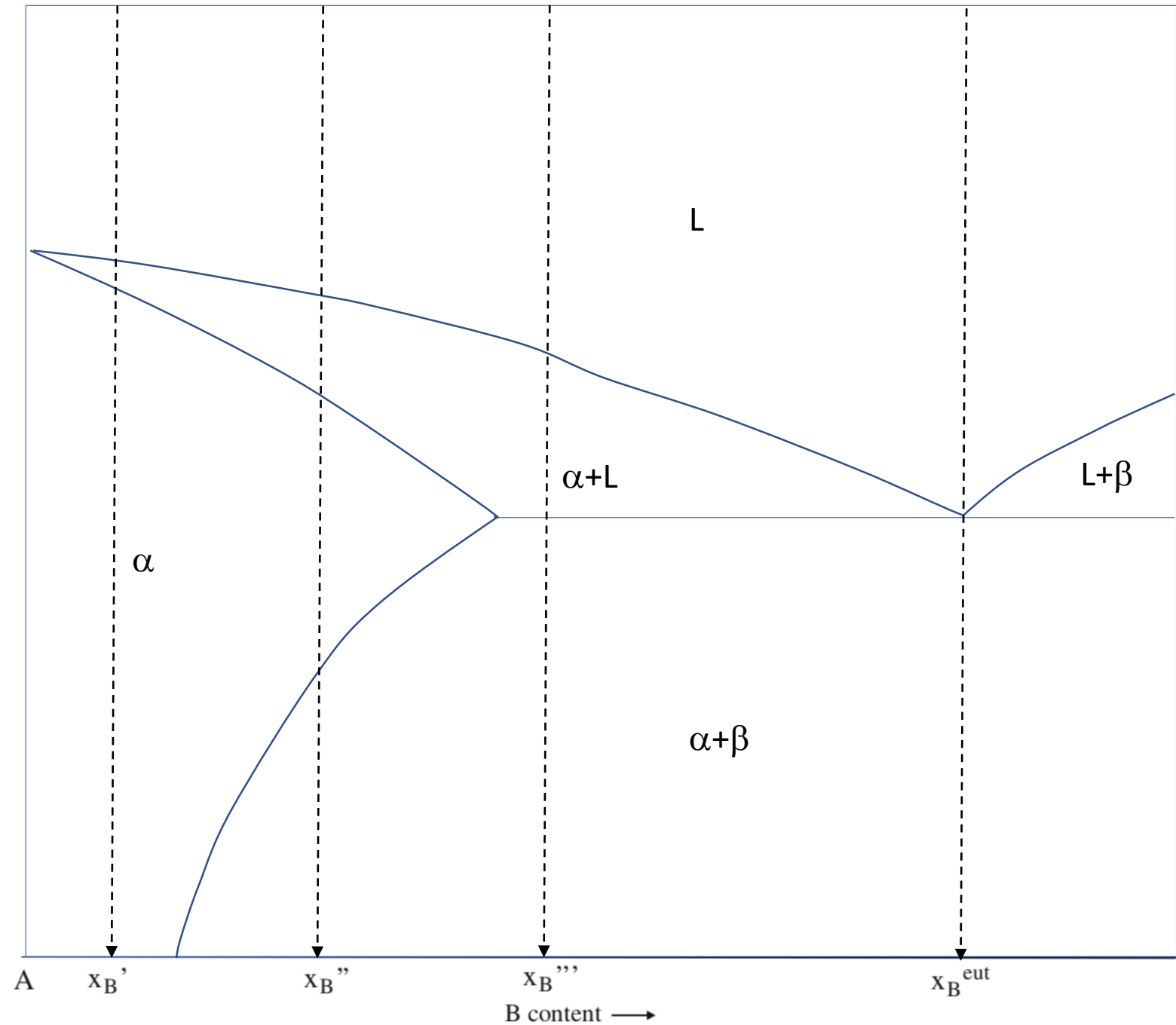


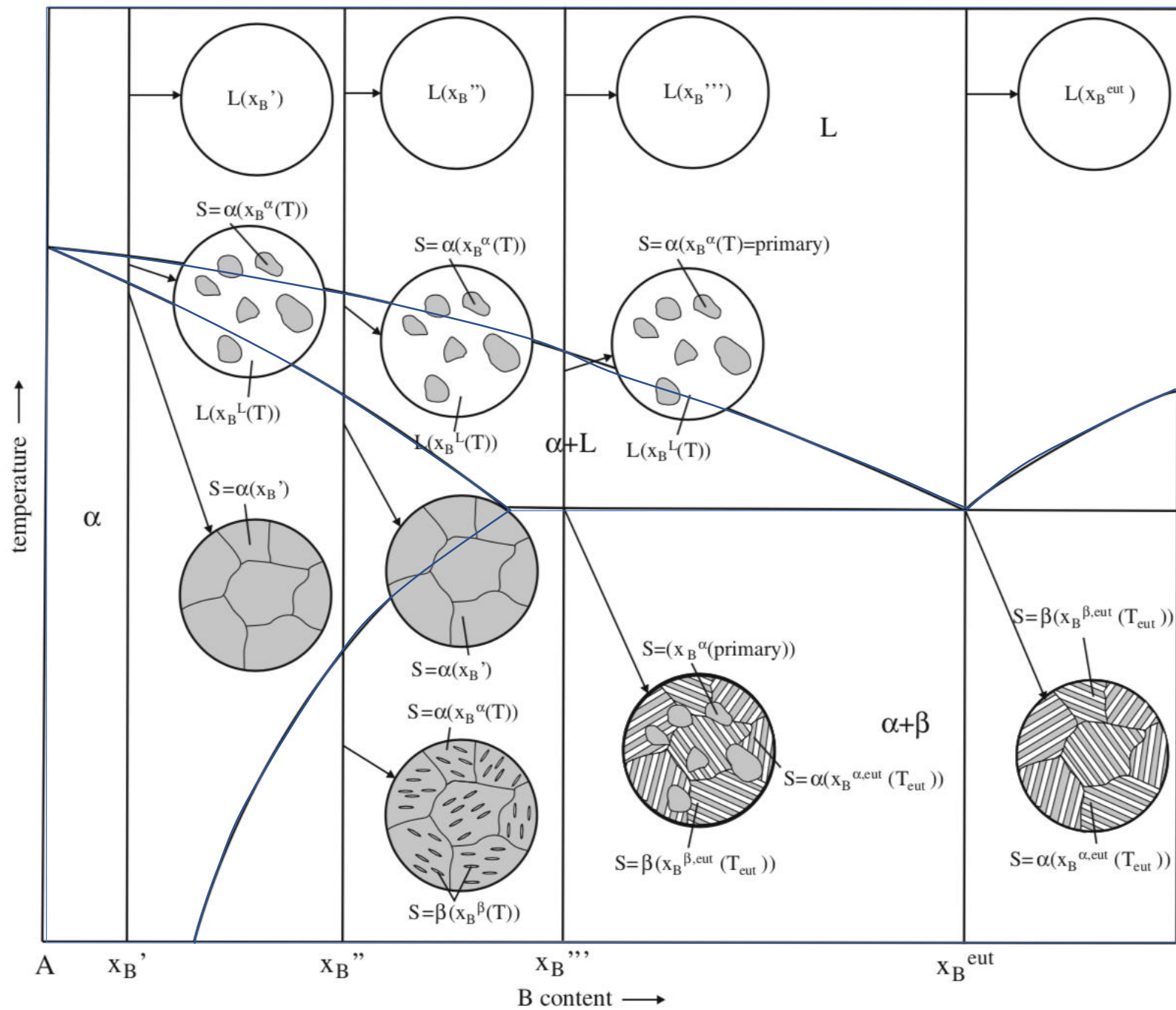
Exercise 3: Draw a phase diagram (for generic components A and B), with complete solubility in the liquid state and **partial solubility in the solid state**. Also indicate the phase/phases present(s) in each region of the phase diagram

Complete solubility in the liquid state,
partial solubility in the solid state

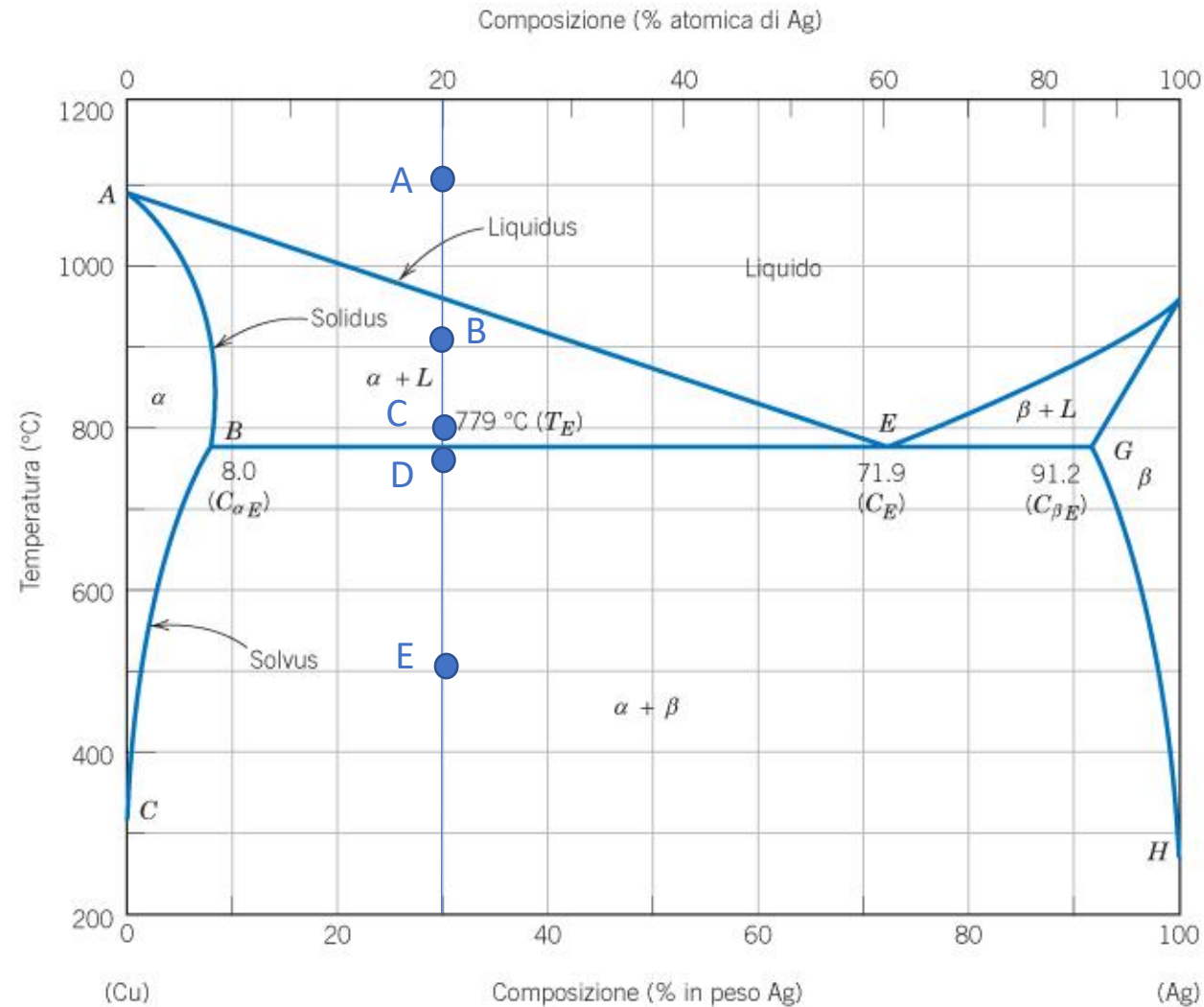


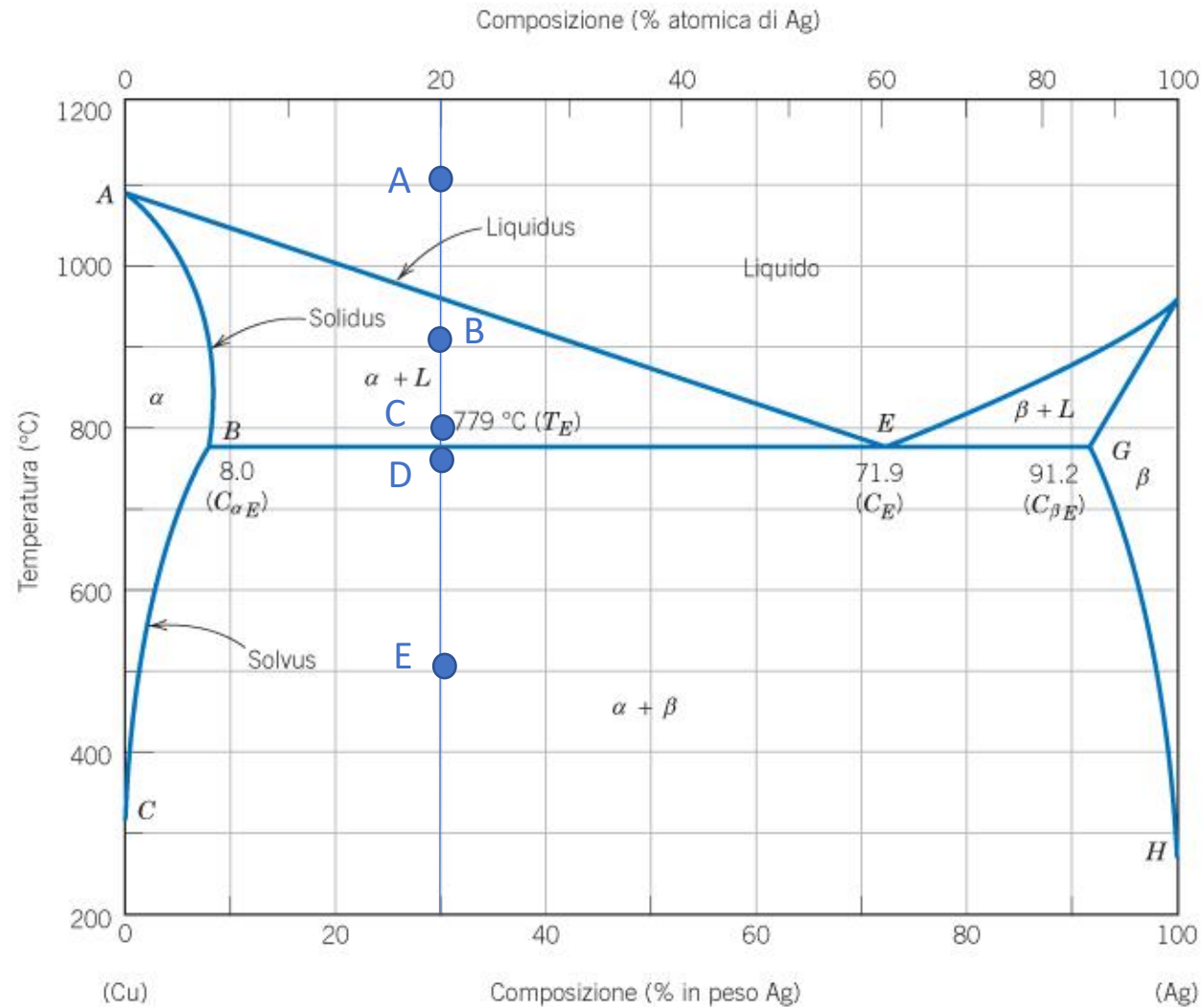
Exercise 4: describe the solidification of alloys with compositions x_B' , x_B'' , x_B''' and x_B^{eut} by drawing schematic microstructures in significant points.





Exercise 5: determine the phase present, their composition and relative amounts in the points **A**, **B**, **C** (at the beginning of the eutectic transformation), **D** (at the end of the eutectic transformation) and **E** indicated in the phase diagram. Schematize the microstructure in each point.



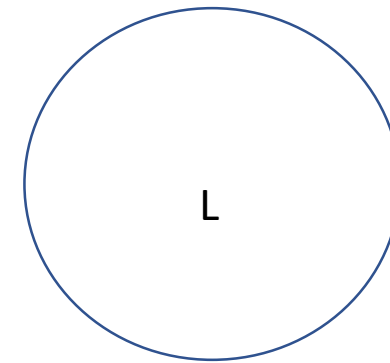


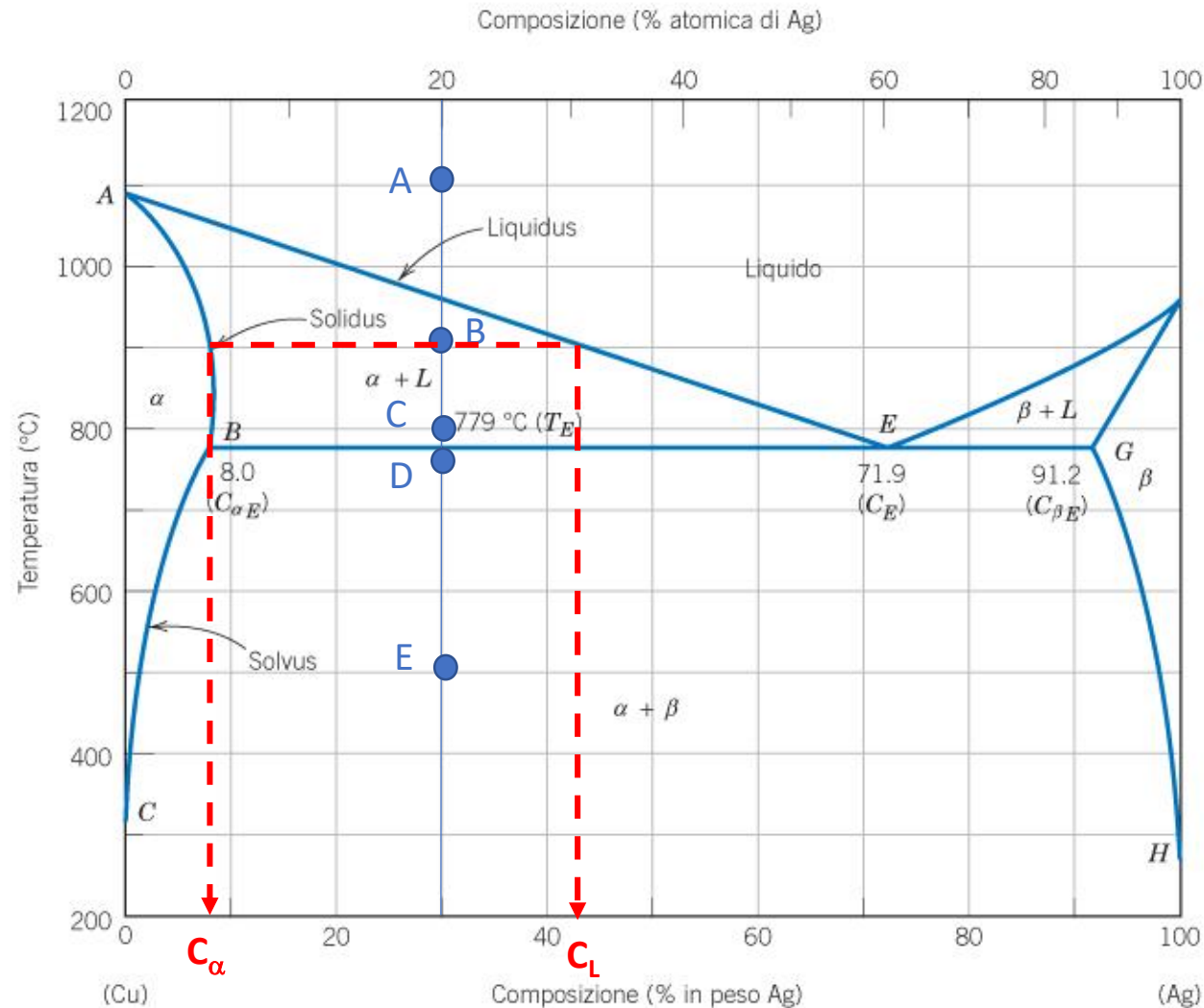
Point A:

Phases: 1 phase liquid

Composition: 30%Ag, 70%Cu

Relative amount: 100% liquid phase





Point B:

Phases: 2 phases solid (α) + liquid (L)

Composition: 2 phases region → horizontal rule

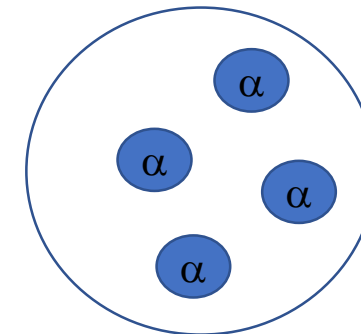
Composition of liquid (L): $C_L = 42\% \text{ Ag}$, $58\% \text{ Cu}$

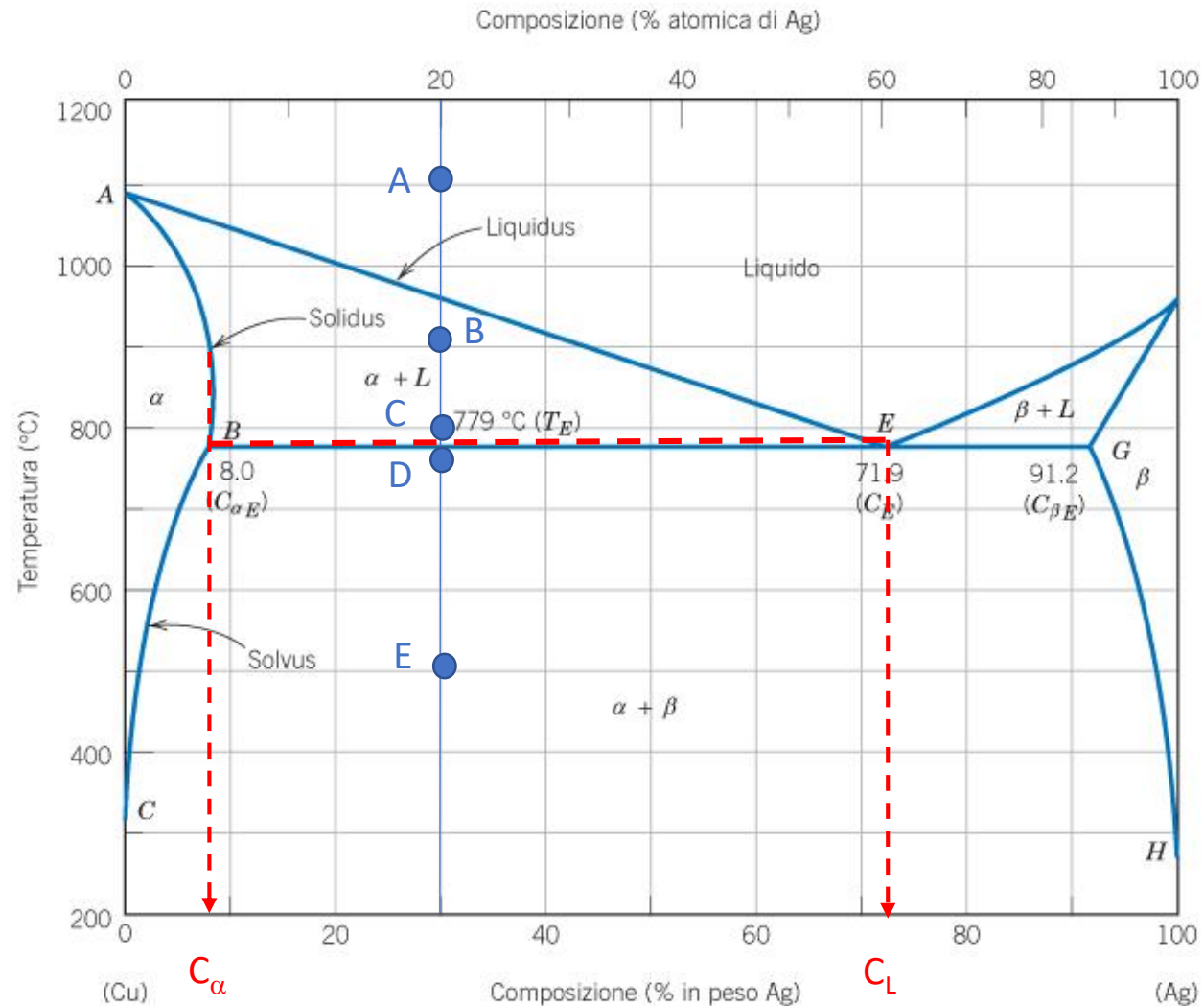
Composition of solid (α): $C_{\alpha} = 8\% \text{ Ag}$, $92\% \text{ Cu}$

Relative amount: 2 phases region → lever rule

$$\% \alpha = (42 - 30) / (42 - 8) * 100 = 35\%$$

$$\% L = (30 - 8) / (42 - 8) * 100 = 65\%$$





Point C:

Phases: 2 phases solid (α) + liquid (L)

Composition: 2 phases region → horizontal rule

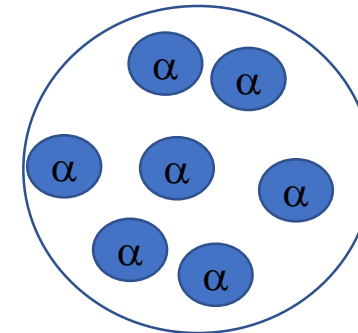
Composition of liquid (L): $C_L = 71.9\% \text{ Ag}$, $28.1\% \text{ Cu}$

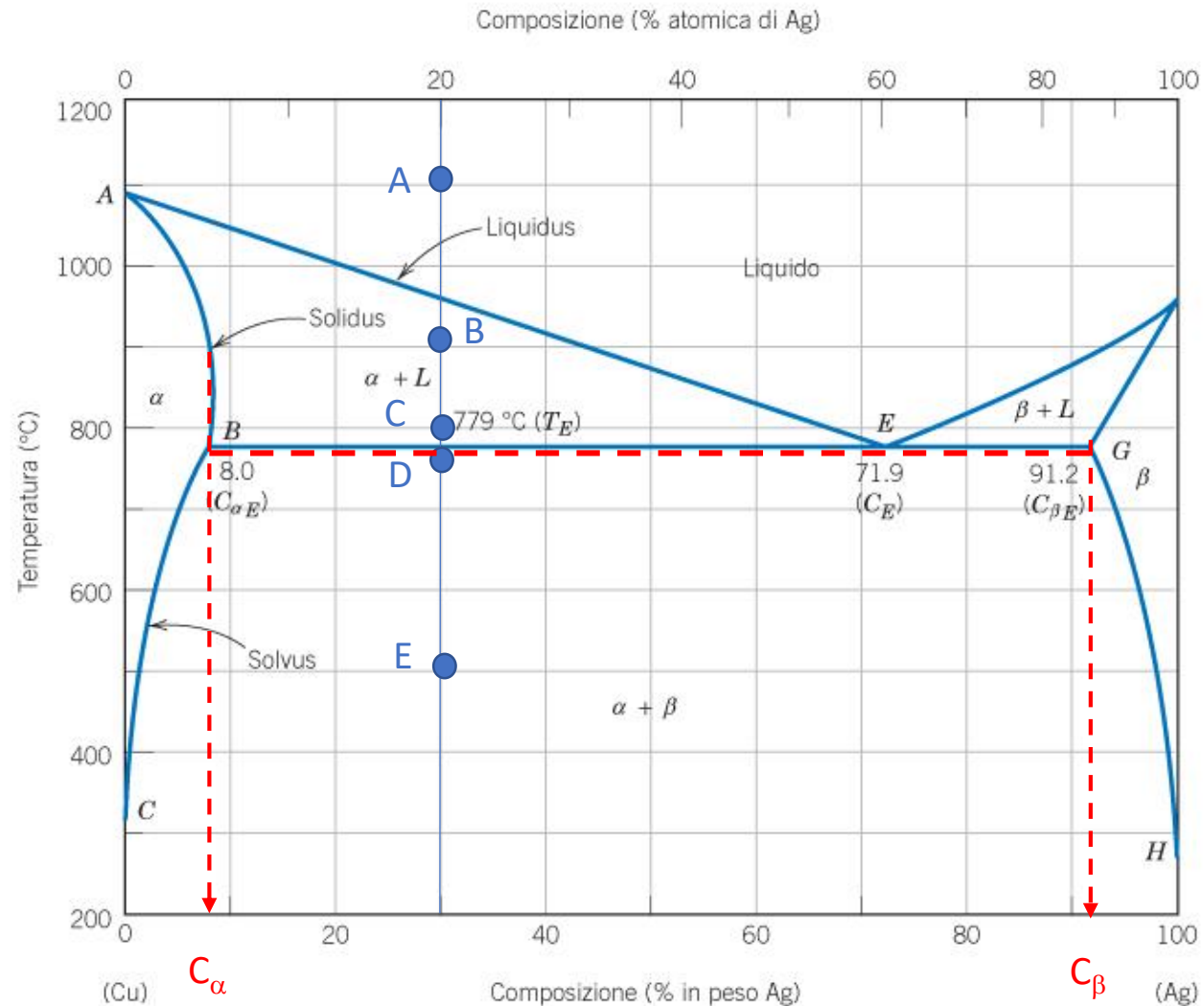
Composition of solid (α): $C_\alpha = 8\% \text{ Ag}$, $92\% \text{ Cu}$

Relative amount: 2 phases region → lever rule

$$\% \alpha = (71.9 - 30) / (71.9 - 8) * 100 = 65.6\%$$

$$\% L = (30 - 8) / (71.9 - 8) * 100 = 34.4\%$$





Point D:

Phases: 2 phases solid (α) + solid (β)

Composition: 2 phases region $\rightarrow$ horizontal rule

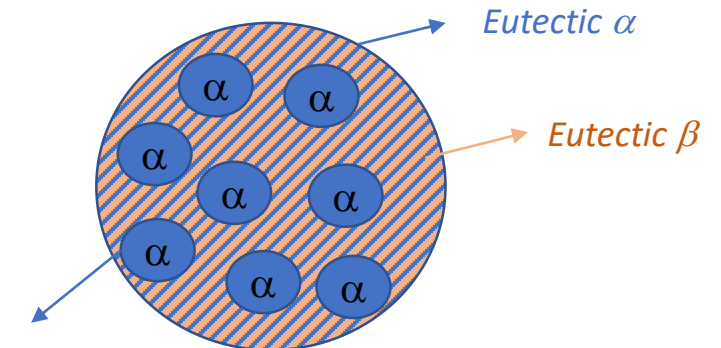
Composition of solid (α): $C_{\alpha} = 8\%\text{Ag}, 92\%\text{Cu}$

Composition of solid (β): $C_{\beta} = 91.2\%\text{Ag}, 8.8\%\text{Cu}$

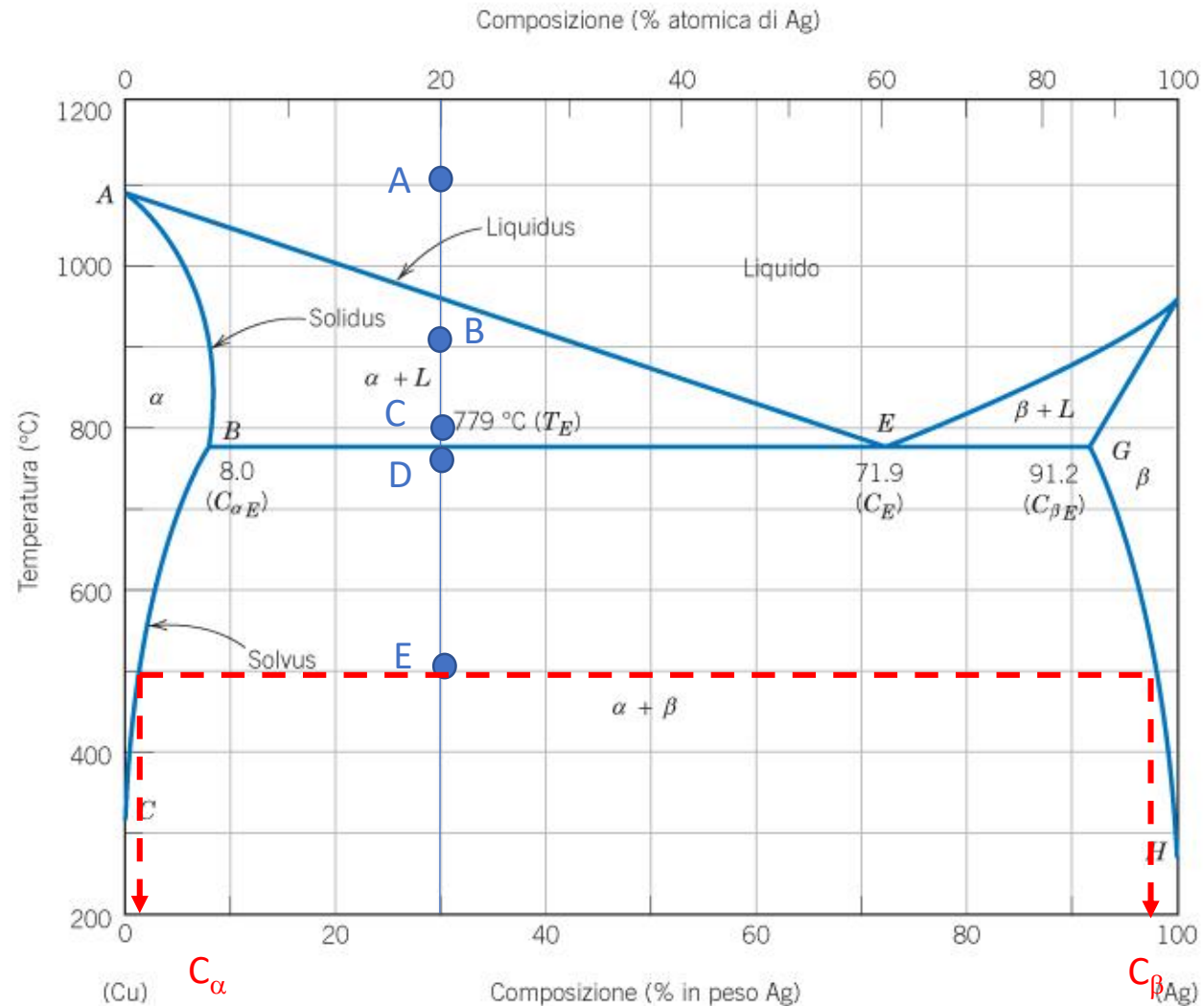
Relative amount: 2 phases region $\rightarrow$ lever rule

$$\% \alpha = (91.2 - 30) / (91.2 - 8) * 100 = 73.5\%$$

$$\% \beta = (30 - 8) / (91.2 - 8) * 100 = 26.5\%$$



Primary/pro-eutectic α



Point D:

Phases: 2 phases solid (α) + solid (β)

Composition: 2 phases region → horizontal rule

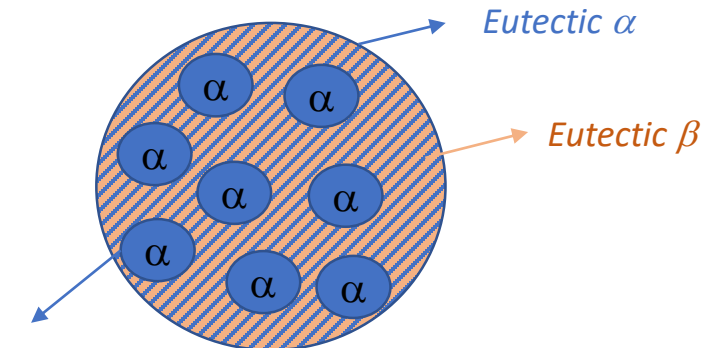
Composition of solid (α): $C_{\alpha} = 2\% \text{Ag}, 98\% \text{Cu}$

Composition of solid (β): $C_{\beta} = 98\% \text{Ag}, 2\% \text{Cu}$

Relative amount: 2 phases region → lever rule

$$\% \alpha = (98 - 30) / (98 - 2) * 100 = 71\%$$

$$\% \beta = (30 - 2) / (98 - 2) * 100 = 29\%$$



Eutectic phase diagram

